Electronic Supplementary Material for "Do Visual-Selection Heuristics Measure Technical Quality? A Controlled Audit of Sixteen Operations Used in Video Summarization"

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This supplement holds the detailed breakdowns behind results reported in the article, relocated here to keep the article at a readable length. Nothing in it is new: every figure quoted in the article is reproduced in full below, and the article cites these tables as Table S1, Table S2, and so on.

Table S1 In-domain collection: topics of Brazilian television news, with the number of source videos, segment slots, redundancy clusters and segment duration range. *Slots* counts cluster memberships; because two segments belong to two clusters each, the 33 slots correspond to 31 distinct segments.

Topic	Videos	Slots	Clusters	Duration (s)	Status
bebe_real	3	5	2	8–41	Included
brumadinho	3	6	3	13–188	Included
fone_canc._ruido	3	6	3	16–30	Included
isencao_imposto	3	6	3	25–30	Included
mounjaro	3	4	1	8–25	Included
ucrania	3	6	3	25–29	Included
esquema_pcc	3	–	–	–	Excluded[†]
Total	**18**	**33**	**15**	8–188	—

[†]98 SIFT descriptors $< K = 300$ K-means centroids. Slots exceed distinct segments in brumadinho and isencao_imposto, where one segment belongs to two clusters.

Table S1 (continued). Operational 42-cluster set used in the primary application analysis.

Topic	Clusters	Slots	Distinct segments
acordo_mariana	1	2	2
aumento_cafe	1	4	4
bebe_real	2	5	5
bolsonaro_trump	3	6	6
brasil_omc	1	2	2
brumadinho	3	6	5
catedral_notre_dame	2	5	4
cessar_fogo_ucrania	3	6	4
esquema_pcc	3	6	6
fone_cancelamento_ruido	3	6	6
fraude_inss	3	6	4
google_huawei	3	6	4
isencao_imposto	3	6	5
julgamento_bolsonaro	3	6	5
mounjaro	1	4	4
recorde_calor_2024	2	5	5
rio_celular	1	2	2
ucrania	3	6	6
ultraprocessados	1	4	4
Total	**42**	**93**	**83**

Table S2 Illustration of the sensitivity axis on a single broadcast frame and four degradations, after applying each method's declared polarity. The pattern this small case shows is the one Table S3 then measures over 25 distortion types and 10,125 images; nothing is concluded from this table alone. Cells are shaded by outcome: expected order, partial, violated.

Method	Pol.	Original	Blur	Noise	Dark	JPEG	Outcome
QualiCLIP	↑	0.341	0.155	0.167	0.196	0.163	as expected
LIQE	↑	1.294	1.056	1.000	1.023	1.023	as expected
BRISQUE	↓	42.8	90.0	66.4	59.1	78.2	as expected
NIQE	↓	4.53	11.99	15.79	6.18	9.47	as expected
Keypoint density	↑	2598	104	**4543**	4	2799	partial
DINOv2 ViT-S/14	↑	14.69	14.86	14.86	14.90	15.14	violated
ResNet-50	↑	15.62	**18.11**	15.00	15.97	15.28	violated

Values in bold are the ones that violate the expected order.

Table S2 (continued). Exact parameters of the constructed broadcast degradations.

Type	Levels 1–5
Gaussian blur	Kernel width $\{3, 5, 9, 15, 23\}$ pixels
White noise	Gaussian $\sigma = \{5, 12, 22, 35, 50\}$; deterministic seeds 1001–1005
Darkening	RGB multiplier $\{.80, .62, .46, .32, .20\}$ with clipping
JPEG	OpenCV quality $\{40, 25, 15, 9, 5\}$
H.264	libx264 CRF $\{30, 36, 41, 45, 50\}$, preset `veryfast`
Rescale	Downscale factors $\{.75, .55, .40, .28, .18\}$, then restore original size
Motion blur	Horizontal kernel width $\{5, 9, 15, 23, 33\}$ pixels
Desaturation	HSV saturation multiplier $\{.75, .55, .35, .18, .05\}$

Each segment contributes at most 30 uniformly spaced frames. The seeded 20-segment simple random sample represents 13 topics: one segment each from `aumento_cafe`, `bolsonaro_trump`, `cessar_fogo_ucrania`, `esquema_pcc`, `isencao_imposto`, `mounjaro` and `rio_celular`; two each from `bebe_real`, `google_huawei`, `julgamento_bolsonaro`, `recorde_calor_2024` and `ucrania`; and three from `fone_cancelamento_ruido`.

Table S3 Sensitivity to distortion level on KADID-10k. Direction-aligned SRCC is computed within each of the 81 reference images across five levels and then averaged by distortion type and across the 25 types. Intervals block-bootstrap the 81 references (10^4 resamples, seed 42). The last column counts types whose reference-averaged SRCC exceeds 0.50.

Method	Mean ρ^*	95% block CI	Types > 0.5
Q-Align	**0.880**	[0.874, 0.886]	**23/25**
QualiCLIP	**0.849**	[0.840, 0.857]	**22/25**
CLIP-IQA+	**0.808**	[0.795, 0.821]	**22/25**
LIQE	**0.807**	[0.797, 0.817]	**22/25**
MUSIQ	**0.733**	[0.720, 0.747]	**20/25**
TOPIQ	**0.721**	[0.702, 0.737]	**20/25**
HyperIQA	0.673	[0.653, 0.692]	**19/25**
DBCNN	0.668	[0.642, 0.692]	**19/25**
ARNIQA†	0.668	[0.652, 0.683]	**20/25**
BRISQUE	0.657	[0.628, 0.684]	**19/25**
CLIP-IQA	0.645	[0.612, 0.676]	**20/25**
TReS†	0.620	[0.598, 0.640]	**18/25**
NIQE	0.607	[0.587, 0.625]	**19/25**
SI	0.251	[0.219, 0.282]	13/25
Keypoint density	0.079	[0.033, 0.128]	7/25
SIFT+BoVW	−0.077	[−0.124, −0.030]	5/25
DINOv2	−0.064	[−0.136, 0.009]	1/25
ResNet-50	−0.297	[−0.364, −0.225]	0/25

†ARNIQA and TReS are reported on this axis as coverage extensions but are not treated as complete-case methods in the screening rule because the required CLIVE evaluations were not available.

Table S4 P.1401 point report after the logistic mapping. PLCC measures linearity, RMSE accuracy and the outlier ratio (OR; lower is better) consistency; RMSE* is the companion error after discounting the 95% confidence interval of the subjective score. OR saturates on KonIQ-10k because its per-image subjective intervals are narrow, but remains discriminative on KADID-10k. Analytic 95% intervals for PLCC, RMSE, RMSE* and OR are included in the released machine-readable results. A dagger marks evaluation on training data.

Method	KonIQ-10k				KADID-10k			
	PLCC	RMSE	RMSE*	OR	PLCC	RMSE	RMSE*	OR
Q-Align†	.940	5.251	5.166	.983	.952	.330	.155	.292
LIQE†	.915	6.234	6.149	.983	.720	.751	.526	.703
QualiCLIP	.771	9.837	9.752	.990	.673	.800	.572	.725
TOPIQ†	.978	3.198	3.124	.959	.546	.907	.671	.769
HyperIQA†	.956	4.507	4.428	.977	.503	.936	.700	.776
DBCNN†	.935	5.483	5.399	.982	.499	.939	.700	.799
MUSIQ†	.904	6.612	6.527	.985	.596	.869	.641	.730
CLIP-IQA+†	.863	7.790	7.705	.987	.665	.809	.581	.727
CLIP-IQA	.724	10.647	10.561	.992	.521	.924	.691	.787
SI	.561	12.782	12.696	.993	.328	1.023	.783	.821
SIFT+BoVW (canon.)	.484	13.508	13.423	.995	.224	1.055	.814	.835
Keypoint density	.484	13.502	13.417	.994	.227	1.055	.814	.836
NIQE	.393	14.194	14.108	.993	.439	.973	.734	.807
BRISQUE	.253	14.931	14.844	.995	.395	.995	.758	.807
ResNet-50	.162	15.231	15.143	.994	.194	1.062	.821	.834
DINOv2	.065	15.402	15.315	.996	.041	1.082	.841	.840

Table S5 Mean direction-aligned SRCC by paradigm. The two blocks are separated because the change between databases means different things in each. In the upper block no member was fitted on either database, so Δ compares two generalization results. In the lower block at least one member was fitted on KonIQ-10k, marked †; there the KonIQ-10k column mixes fit with generalization and no Δ is meaningful.

Paradigm	*N*	KADID	KonIQ	Δ	Best member
Neither database seen by any member					
ITU-T complexity	1	0.219	0.513	+0.294	SI
SIFT+BoVW*	2	0.124	0.446	+0.322	canonical IDF
Handcrafted	1	0.123	0.447	+0.324	Keypoint density
NR-IQA, NSS	2	0.355	0.302	−0.053	NIQE
CNN+BoVW	19	0.318	0.058	−0.260	VGG-19+BoVW
Deep features	2	−0.086	−0.115	−0.029	DINOv2 ViT-S/14
At least one member fitted on KonIQ-10k					
NR-IQA, LMM	1	**0.951**†	**0.920**†	—	Q-Align
NR-IQA, CNN reg.	2	0.473	**0.933**†	—	HyperIQA
NR-IQA, transformer	2	0.534	**0.923**†	—	TOPIQ
NR-IQA, CLIP	4	0.636	**0.774**†	—	LIQE

*Mean over the magnitudes of the two TF-IDF variants, which differ in orientation but not in strength. Restricted to the two CLIP members fitted on neither database, that paradigm gives 0.583 on KADID-10k and 0.679 on KonIQ-10k, a change of +0.096.

Table S6 The BoVW variants in the domain, on the agreement axis and on what they deliver. *Differs* counts the clusters in which the variant selects a different segment from the deployed one; *Agreement* is top-1 agreement with the reference model over the 15 clusters; r_{kp} is the rank correlation with keypoint density over the 33 segment slots; and the last three columns are the summary the variant delivers, as in Table S10. The comparison between summing and averaging defined in the main article is therefore evaluated directly. The final column is the uncalibrated raw sum of the DOVER technical and aesthetic branches, used only as a scale-sensitivity diagnostic.

Variant	Aggr.	Differs	Agreem.	r_{kp}	Dur. (s)	Q-Align	raw T+A
SIFT+BoVW (as implemented)	sum	—	9/15	+0.591	634.0	61.1%	52.1%
SIFT+BoVW (canonical IDF)	sum	1/15	10/15	+0.116	801.5	62.2%	47.6%
BoVW / descriptor count	mean	2/15	10/15	−0.199	810.9	59.0%	48.2%
BoVW / frame count	mean	1/15	10/15	−0.145	**459.3**	**77.3%**	**71.5%**
Keypoint density	mean	5/15	7/15	+1.000	—	—	—
LIQE (for reference)	mean	5/15	11/15	−0.381	455.6	85.8%	91.8%

Table S7 Descriptive in-domain agreement with the Q-Align reference model over the 15 human-annotated redundancy clusters, raw and chance-corrected against $\bar{p}_e = 0.472$. These clusters share topics and two pairs share a segment; the table therefore makes no cluster-independence inference. Topic-block intervals for the larger system-defined set are reported in the main article. Roles follow Table 1: D = deployed, C = candidate and X = diagnostic.

Method	Role	Agreement	$\kappa_{\mathrm{uniform}}$
MUSIQ	C	13/15 (86.7%)	+0.747
DBCNN	C	12/15 (80.0%)	+0.621
TOPIQ	C	12/15 (80.0%)	+0.621
LIQE	C	11/15 (73.3%)	+0.495
HyperIQA	C	11/15 (73.3%)	+0.495
NIQE	C	10/15 (66.7%)	+0.368
QualiCLIP	C	10/15 (66.7%)	+0.368
BRISQUE	C	10/15 (66.7%)	+0.368
SIFT+BoVW	D	9/15 (60.0%)	+0.242
CLIP-IQA	C	8/15 (53.3%)	+0.116
CLIP-IQA+	C	8/15 (53.3%)	+0.116
DINOv2	X	5/15 (33.3%)	−0.263

Table S8 The four redundancy clusters in which the method under study diverges from the majority of the quality references. Each row is a candidate segment; each column a scoring method. The top-1 choice of each method is shaded: pink for SIFT+BoVW, green for the quality references. NIQE and BRISQUE are lower-is-better. Scores are reproduced from the aggregated per-cluster results.

Segment	SIFT+BoVW	LIQE	NIQE↓	Q-Align	QualiCLIP	BRISQUE↓
bebe_real C1						
Band 0:37 (10 s)	**5448.0**	1.49	6.55	4.02	0.28	50.97
Record 0:46 (11 s)	2077.6	**1.59**	**5.57**	**4.17**	**0.30**	**41.44**
brumadinho C2						
Band 0:00 (13 s)	2888.3	**2.60**	**5.04**	**4.84**	**0.36**	**44.95**
Record 4:24 (188 s)	**5332.7**	1.13	6.23	3.74	0.27	53.57
isencao_imposto C0						
Band 2:46 (30 s)	**3732.8**	4.23	**4.67**	3.70	**0.62**	34.65
CNN 1:12 (25 s)	3010.0	**4.47**	4.74	**4.26**	0.59	**29.84**
mounjaro C0						
Band 0:00 (8 s)	559.8	4.12	7.29	**4.38**	0.56	63.19
Dom. Esp. 2:06 (16 s)	**1945.1**	3.95	5.87	3.90	0.57	36.55
Dom. Esp. 3:07 (25 s)	1784.4	3.61	5.28	3.68	0.56	42.29
Record 0:00 (14 s)	815.2	**4.93**	**4.76**	4.26	**0.63**	**35.45**

Table S9 Eligible set as a function of the generalization threshold, under both readings of "a database it was not fitted on". *Universal* requires the threshold to hold on every unseen database (the reading adopted in the main article); *existential* requires it on at least one. The sensitivity requirement is held at a within-reference mean above 0.50 and at least 15/25 types; the topic-block lower confidence limit must also exceed the in-domain floor.

Minimum ρ^*	Universal reading (adopted)	Existential reading
0.50	LIQE, QualiCLIP, CLIP-IQA+, MUSIQ, TOPIQ, CLIP-IQA	LIQE, QualiCLIP, CLIP-IQA+, MUSIQ, TOPIQ, CLIP-IQA, DBCNN, HyperIQA
0.55	LIQE, QualiCLIP, CLIP-IQA+, MUSIQ	*same eight methods*
0.60	**LIQE, QualiCLIP, CLIP-IQA+**	*same eight methods*
0.65*	**LIQE, QualiCLIP, CLIP-IQA+**	*same eight methods*
0.70	LIQE	*same eight methods*
0.75	none	LIQE, QualiCLIP, CLIP-IQA+, MUSIQ, TOPIQ, DBCNN, HyperIQA

*The value used. At the adopted generalization threshold and universal reading, the same three methods survive for sensitivity type-count cuts from 12/25 through 22/25; none survives at 23/25. Requiring only a positive floor point estimate instead of a topic-block lower limit does not change the three-method set.

Table S10 The same comparison on the 15 human-annotated clusters. The two references agree closely here; their disagreement emerges on the 42-cluster analysis in the main article, which is the primary estimate. The DOVER column uses the uncalibrated raw technical-plus-aesthetic sum, not DOVER-T.

Scoring method	Changed	by raw T+A	by Q-Align	Dur. (s)	Words
SIFT+BoVW (deployed)	—	52.1%	61.1%	634.0	1,613
LIQE	5/15	91.8%	85.8%	455.6	1,124
BRISQUE	6/15	84.5%	84.7%	431.0	1,073
QualiCLIP	4/15	80.9%	80.4%	439.6	1,124
NIQE	4/15	80.9%	80.4%	439.6	1,124
DINOv2	12/15	53.4%	36.7%	328.6	904

Table S11 (continued). Construct and scale sensitivity on the 42 operational clusters.

DOVER output	Estimand	SIFT	LIQE	Δ; topic CI	Hierarchical CI
Technical branch	Global range (%)	50.5	76.0	+25.5; [−12.1, 54.9]	[−17.2, 59.6]
Technical branch	Raw mean	−.0770	−.0701	+.0069; [−.0028, .0178]	[−.0041, .0197]
Technical branch	Mean cluster position (%)	57.1	64.4	+7.2; [−20.7, 35.1]	[−22.4, 37.5]
Technical branch	Selected maximum (%)	54.8	59.5	+4.8; [−22.2, 31.7]	[−24.4, 34.2]
Raw T+A sum	Global range (%)	47.4	77.7	+30.3; [1.1, 54.1]	[−4.1, 59.0]
Aesthetic branch	Global range (%)	45.0	73.7	+28.7; [5.3, 49.7]	[−.2, 54.6]

The branch sum is the raw technical output plus the raw aesthetic output. It is not the official ITU-standardized DOVER fused score. The technical branch is primary; the other rows diagnose construct sensitivity.

DOVER-T range filter	Clusters	Threshold	Global-range Δ
All clusters	42	—	+25.5
Exclude bottom 10%	37	.00397	+25.9
Exclude bottom 25%	31	.00891	+28.1

DOVER-T cluster ranges: minimum .00028, first quartile .00891, median .01686, third quartile .04118, maximum .10653; zero-range clusters: 0.

Matched-input control: ten identical frames and mean pooling.

Scorer	Differs from SIFT	DOVER-T range	Raw mean	Selected max
SIFT+BoVW	—	52.0%	−.0766	57.1%
LIQE	16/42	79.1%	−.0692	64.3%

LIQE–SIFT global-range Δ = +27.1 points (topic CI [5.6, 47.8]; hierarchical CI [2.4, 52.5]). Raw-branch Δ = +.00736 (topic [.00135, .01406]; hierarchical [.00057, .01683]). Thirteen of the 16 disagreement clusters belong to the 22-cluster complete-operation change set. Relative to the published deployed path, matched-input SIFT changes 12 choices and matched-input LIQE changes 22; the control isolates scorer identity, not the historical operation itself.

The complete registered commit is:

`9377c07ca080ac61c7cff80e0ff6c7507b6ebaba`.

The submission archive contains `registered_prediction_commit.bundle`; the clone, bundle verification and object-inspection commands in its README make the commit and timestamp independently inspectable even before the public release.

Table S11 The eight predictions registered at commit `9377c07` before the constructed experiment, against the measurements. None is used to filter a method.

	Subject	Prediction	Measured	Outcome
P1	SIFT+BoVW	High accuracy on blur and darkening	90%, 100%	holds
P2	SIFT+BoVW	Inverted on white noise	8%	holds
P3	SIFT+BoVW	Near chance on JPEG and H.264	6% and 90%	refuted
P4	CNN+BoVW (19)	Chance on every type	44.9%, CI [41.0, 48.9]; range 30–64	refuted
P5	NIQE, BRISQUE	Accuracy falls under heavy overlay	+2.0, +2.2 points	refuted
P6	Learned methods	No interaction with overlay	−18.1 to −8.1 for three	refuted
P7	SIFT+BoVW	Design B |mean Spearman| below 0.15 (insensitivity)	−0.425	refuted
P8	Q-Align	Design A accuracy above 0.90	98.2%, CI [97.2, 99.1]	holds

Table S12 (continued). Fidelity and descriptor-count diagnostics for the constructed broadcast pairs.

Method	All 8 types (800)			Without dark/desaturation (600)		
	Correct	Ties	Accuracy	Correct	Ties	Accuracy
SIFT+BoVW	521	0	65.1%	382	0	63.7%
LIQE	796	0	99.5%	599	0	99.8%
Q-Align	786	0	98.2%	600	0	100.0%
DOVER-T	781	0	97.6%	599	0	99.8%

Type	Mean ΔD	$\Delta D > 0$	Mean ΔD by level 1–5
Blur	−797	10%	−130, −417, −913, −1188, −1339
Dark	−796	0%	−229, −487, −791, −1116, −1358
H.264	−398	10%	−128, −220, −361, −541, −737
JPEG	+289	95%	+161, +252, +339, +376, +317
Motion blur	−1067	0%	−548, −909, −1174, −1308, −1396
Noise	+469	92%	+126, +348, +517, +607, +749
Rescale	−646	12%	−154, −377, −585, −911, −1204
Desaturation	+18	61%	0, +14, +21, +34, +24

$\Delta D = D_{\text{degraded}} - D_{\text{unmodified}}$. Strict accuracy uses correct/total; a tie is neither correct nor incorrect and remains in the denominator.

DOVER-T	dark	motion	blur	H.264	rescale	satur.	noise	JPEG	All
Accuracy (%)	93	100	100	100	100	89	99	100	97.6
Mean severity ρ^*	.985	.900	1.000	1.000	1.000	.715	.995	1.000	—

The raw technical output is the exact branch later used for the 42-cluster output analysis. Its overall segment-block 95% CI is [96.1, 99.0]; after excluding darkening and desaturation, accuracy is 99.8% with CI [99.5, 100]. There are no ties. Each condition is encoded through the same 5-fps MP4 wrapper and uses paired spatial fragment seed 42.

Table S12 Method inventory and released-weight supervision. *Sup.*: OU = opinion-unaware, K = KonIQ-10k, L = LIVE, M = multi-database mixture. Roles follow Table 1: D = deployed, C = candidate, R = reference and X = diagnostic or ablation. The daggered partial-coverage methods are excluded from complete-case screening.

Method	Family	Construct	Sup.	Pol.	GPU	Role
SIFT+BoVW	Handcrafted	TF-IDF word count	–	↑	No	D
Keypoint density	Handcrafted	keypoints/frame	–	↑	No	X
DINOv2	Self-supervised	patch-token norm	–	↑	Yes	X
ResNet-50	Supervised	pooled-feature norm	–	↑	Yes	X
BRISQUE	NSS	natural-statistics deviation	L	↓	No	C
NIQE	NSS	pristine-model distance	OU	↓	No	C
QualiCLIP	CLIP	quality score	OU	↑	Yes	C
LIQE	CLIP	quality, distortion, scene	K	↑	Yes	C
Q-Align	LMM	textual quality levels	M	↑	Yes	R
SI	ITU-T complexity	Sobel energy	–	↑	No	X
19 CNN+BoVW	ablation	TF-IDF word count	–	↑	Yes	X
DBCNN	CNN regression	bilinear pooling	K	↑	Yes	C
HyperIQA	CNN regression	adaptive weights	K	↑	Yes	C
MUSIQ	Transformer	multi-scale score	K	↑	Yes	C
TOPIQ	Transformer	semantic attention	K	↑	Yes	C
CLIP-IQA	CLIP	antonym prompts	OU	↑	Yes	C
CLIP-IQA+	CLIP	tuned prompts	K	↑	Yes	C
ARNIQA†	Self-supervised	distortion manifold	K	↑	Yes	partial
TReS†	CNN–transformer	ranking/consistency	K	↑	Yes	partial

PyIQA registry keys used by the archived scorers are `brisque`, `niqe`, `dbcnn`, `hyperiqa`, `musiq`, `topiq_nr`, `clipiqa`, `clipiqa+`, `liqe`, `qualiclip`, `arniqa` and `tres`. The final inventory and analysis archive were frozen on 23 August 2026. MANIQA was attempted with PyIQA 0.1.15.post2 and the `maniqa-kadid` checkpoint. A single-image diagnostic completed, but the released implementation evaluates 20 crops per image and the full 21,360-image three-database run did not finish within the experiment window on the single 8-GB RTX 4060; it is therefore an uncompleted method, not a failed candidate.

Table S12 (continued). Complete extractor inventory for the CNN+BoVW ablation. Every row uses the final spatial tensor before global pooling or classification.

Extractor	Released torchvision weights	Spatial endpoint
ResNet-18	IMAGENET1K_V1	trunk before average pooling
ResNet-50	IMAGENET1K_V2	trunk before average pooling
ResNet-101	IMAGENET1K_V2	trunk before average pooling
VGG-16	IMAGENET1K_V1	full `features` module
VGG-19	IMAGENET1K_V1	full `features` module
EfficientNet-B0	IMAGENET1K_V1	full `features` module
MobileNet-V3-Large	IMAGENET1K_V2	full `features` module
ConvNeXt-Tiny	IMAGENET1K_V1	full `features` module
ConvNeXt-Small	IMAGENET1K_V1	full `features` module
DenseNet-121	IMAGENET1K_V1	full `features` module
DenseNet-201	IMAGENET1K_V1	full `features` module
ResNeXt-50-32x4d	IMAGENET1K_V2	trunk before average pooling
ResNeXt-101-64x4d	IMAGENET1K_V1	trunk before average pooling
Wide-ResNet-50-2	IMAGENET1K_V2	trunk before average pooling
Wide-ResNet-101-2	IMAGENET1K_V2	trunk before average pooling
Inception-V3	IMAGENET1K_V1	convolutional trunk through Mixed_7c
EfficientNet-B4	IMAGENET1K_V1	full `features` module
SqueezeNet-1.1	IMAGENET1K_V1	full `features` module
MobileNet-V2	IMAGENET1K_V2	full `features` module

Common preprocessing: bicubic resize to 256 pixels, center crop to 224×224, ImageNet mean and standard deviation, and FP16 inference. Each spatial location is flattened into one descriptor and ℓ_2-normalized in FP32. The same internal key-frame filter, vocabulary construction, TF–IDF weighting and segment aggregation are then applied. The archived implementation is `cnn_bovw_scorer.py`.

Table S12 (continued). Exact configurations of the two model-based references used in the broadcast analyses.

Model	Config.	Input / pooling	Output used
Q-Align	`q-future/one-align`; task `quality`, input `image`	Uniform 1-fps frames; arithmetic mean of frame scores	Four-bit NF4 with double quantisation and FP16 compute; higher score is better. The same configuration scores the 800 constructed pairs and supplies the in-domain floor.
DOVER	Official `DOVER.pth` and `dover.yml`, `val-l1080p` view decomposition	Technical: 7×7 spatial fragments of size 32, 32 frames, interval 2, three clips; aesthetic: 224×224, 32 frames, interval 2, one clip	The technical branch is the primary output reference and is also evaluated on the 800 pairs with a common 5-fps MP4 wrapper and paired seed 42. The raw branch sum is only a sensitivity analysis; it is not the official standardised fusion or DOVER++.

Table S13 Computational cost measured on one RTX 4060 over the same inputs. Frame-wise entries report median latency after ten warm-up and sixty timed iterations; the final two rows are measured complete paths.

Method	ms/frame	IQR	Per cluster
SI	2.2	0.14	0.2 s
QualiCLIP	14.3	1.68	1.2 s
TOPIQ	18.7	0.36	1.5 s
BRISQUE	19.6	0.98	1.6 s
CLIP-IQA+	21.4	0.65	1.8 s
DBCNN	21.5	0.46	1.8 s
LIQE	22.7	0.82	1.9 s
CLIP-IQA	25.2	0.51	2.1 s
NIQE	34.6	0.90	2.8 s
MUSIQ	40.0	9.36	3.3 s
ResNet-50	46.2	8.44	3.8 s
DINOv2	46.4	36.63	3.8 s
Keypoint density	46.8	1.48	3.8 s
HyperIQA	50.1	0.77	4.1 s
SIFT+BoVW	$\sim$ 1,724	—	182 s
Q-Align	322.6	—	26 s

The uniform 1-fps candidate cache contains 2,892 frames from 83 distinct segments. Counting repeated memberships across the 93 cluster slots gives 3,525 segment-frame instances. Candidate latency was timed per decoded frame; the SIFT+BoVW row is the complete operation divided by its input-frame count and therefore includes key-frame selection, descriptor extraction, vocabulary quantisation and aggregation. The per-cluster column is the operational quantity; the per-frame SIFT value is an approximate normalization rather than an identical-path microbenchmark.

Table S14 Descriptive three-database contrasts and released-weight exposure. K–C is KonIQ-10k minus CLIVE; C–KADID is CLIVE minus KADID-10k. Positive values are losses. K = KonIQ-10k opinion training, OU = opinion-unaware and L = LIVE opinion training. The contrasts do not isolate causal effects because the databases, architectures and objectives are unmatched.

Method	Sup.	KonIQ	CLIVE	KADID	K–C	C–KADID
Released checkpoint fitted on KonIQ-10k						
TOPIQ	K	0.973†	0.811	0.511	+0.162	+0.300
HyperIQA	K	0.947†	0.750	0.462	+0.197	+0.288
DBCNN	K	0.919†	0.756	0.484	+0.163	+0.272
MUSIQ	K	0.873†	0.789	0.556	+0.084	+0.233
CLIP-IQA+	K	0.824†	0.805	0.654	+0.019	+0.151
LIQE	K	0.915†	0.865	0.723	+0.050	+0.142
mean over the six					+0.113	+0.231
Never fitted on KonIQ-10k						
QualiCLIP	OU	0.711	0.757	0.665	−0.046	+0.092
CLIP-IQA	OU	0.647	0.703	0.501	−0.056	+0.202
NIQE	OU	0.368	0.449	0.379	−0.081	+0.070
BRISQUE	L	0.236	0.313	0.330	−0.077	−0.017
mean over the four					−0.065	+0.087

†In-distribution. CLIVE contains 1,162 images and is unseen by every checkpoint in the table. The operational partition assigns 33% of the mean KonIQ-to-KADID gap of the fitted group to K–C and 67% to C–KADID; these percentages describe contrasts and are not causal estimates.

Table S15 Spearman association with human scores before and after conditioning on detectable keypoint density. r_{kp} is the rank correlation between the method score and keypoint density. Differences are descriptive attenuations, not causal fractions of "content density."

Method	KonIQ-10k				KADID-10k			
	r_{kp}	$\lvert\rho\rvert$	$\lvert\rho_{\cdot\mathrm{kp}}\rvert$	Diff.	r_{kp}	$\lvert\rho\rvert$	$\lvert\rho_{\cdot\mathrm{kp}}\rvert$	Diff.
Q-Align	+.44	.920	.901	−.019	+.13	.951	.950	−.001
LIQE	+.43	.915	.894	−.020	+.18	.723	.718	−.005
TOPIQ	+.48	.973	.966	−.007	+.21	.511	.500	−.011
HyperIQA	+.48	.947	.933	−.014	+.17	.462	.451	−.011
DBCNN	+.49	.919	.898	−.021	+.22	.484	.472	−.012
MUSIQ	+.44	.873	.841	−.032	+.20	.556	.546	−.010
QualiCLIP	+.53	.711	.626	−.086	+.29	.665	.663	−.002
CLIP-IQA	+.27	.647	.611	−.036	+.22	.501	.489	−.012
SI	+.77	.513	.296	−.218	+.70	.219	.187	−.032
NIQE	−.63	.368	.125	−.243	−.33	.379	.362	−.018
BRISQUE	−.31	.236	.116	−.120	−.13	.330	.319	−.011
SIFT+BoVW (canonical)	+.998	.447	.016	arith.	+.992	.125	.028	arith.
SIFT+BoVW (deployed)	−.9999	.445	n.d.	arith.	−.9998	.123	n.d.	arith.

The shaded rows are the near-tautological regime: conditioning a score on a variable with which it is almost perfectly correlated removes the variable from itself. For the deployed row the partial coefficient is not reported because the denominator approaches zero and the result is numerically unstable.

Table S16 The sixteen operations audited, each linked to the objective its source states and the role it has there. *Tested reading* is the one this article evaluates: higher score means better technical quality, applied to a single frame without a model. No source is claimed to be wrong about its own objective; what is tested is the reuse. *Inverts* counts degradation types on which the operation ranks the degraded copy higher (Table 2 of the main article). Where two works adopt the same operation the use can differ: Laplacian variance rejects low-light frames alongside grey-scale intensity and a Canny detector in Cakmak et al. [1], and enters $0.7\,z(\text{sharp}) + 0.3\,z(\text{bright})$ in Korolkov [8].

Operation	Objective in the source	Role there	Inverts
Averaged aesthetics (3) [2]	Keyframe interestingness	three aesthetic terms; the source gives no combination, and the mean is built here	1/8
Colourfulness (Hasler) [3]	Perceived colourfulness of a scene	stand-alone perceptual index	1/8
Colourfulness (Ke) [7]	Professional vs. snapshot photograph	feature of a naive Bayes classifier	1/8
Composite of five [10]	Keyframe ranking	unweighted mean of five features	1/8
Contrast [7]	Professional vs. snapshot photograph	feature of a naive Bayes classifier	2/8
Edge distribution [7]	Professional vs. snapshot photograph	feature of a naive Bayes classifier	2/8
Entropy [9]	Keyframe diversity and coverage	SE in segment coverage (Eq. 21)	2/8
GLCM contrast [12]	Keyframe extraction	texture descriptor	1/8
Gradient magnitude [9]	Keyframe diversity and coverage	GM in visual attention (Eq. 20)	3/8
Gradient orientation [9]	Keyframe diversity and coverage	GO in segment diversity (Eq. 22)	4/8
Grey-level dispersion [6]	Static-frame selection	one of four inputs to a cost multiplicative in cluster distance	2/8
Hue count [7]	Professional vs. snapshot photograph	feature of a naive Bayes classifier	5/8
Laplacian variance [1]	Blur rejection in frame filtering	sharpness threshold	1/8
Saliency area [9]	Keyframe diversity and coverage	VS in visual attention (Eq. 20)	6/8
Saturation [4]	Aesthetic scoring	one term among several	3/8
Spatial information [5]	Spatial detail of a test sequence	characterization, not quality	—

Tested reading, identical for all sixteen: the operation is computed on one frame and read as a technical-quality score in which a higher value marks the better of two versions of the same content. Fifteen are closed-form; saliency area requires a learned map in its source and is marked as such.

Table S17 The sixteen visual-selection operations the video-summarization literature uses at the selection stage, reimplemented from the published equations and measured under one protocol. *Sensitivity* is the SRCC against the degradation level within each KADID-10k reference, aligned to the direction the source declares for its own statistic, averaged over the 25 distortion types, with the count of types above .50; *Generalization* is the lowest such correlation on any database the method was not fitted on; *Floor* is chance-corrected agreement on the operational clusters. **The two families use different reference models there**: operations against LIQE, the ten no-reference models against Q-Align, so that column is not a comparison between families. The damage columns share one protocol. The rule admits a method when sensitivity exceeds .5 on at least 15 of 25 types and generalization exceeds .65. **No operation clears either damage gate**: the best sensitivity is .298 and the best generalization among the fifteen evaluated is .139. Spatial information declares no direction, carries no verdict and is excluded from every count. **The fifteen are not fifteen independent measurements**: on KADID-10k their score matrix has rank thirteen, because one colourfulness implementation duplicates another exactly and one operation is the mean of three others. The *Floor* column reports the in-domain agreement against the reference model with the lower end of its 95% interval in brackets, so a pass can be read for how much of it survives the interval: the weakest passing operation in this table has a lower bound of +.011.

Operation	Source	Sens.	Types	Gen.	Floor κ [95% lo]	Elig.
Edge distribution	[7]	.298	9/25	+.083	−.258 [−.484]	no
Gradient orientation	[9]	.291	12/25	−.035	—	no
Spatial information	[5]	.251	13/25	+.218	+.506 [+.280]	—
Contrast	[7]	.185	10/25	+.139	+.371 [+.106]	no
Grey-level dispersion	[6]	.149	11/25	+.096	+.101 [−.157]	no
Hue count	[7]	.147	6/25	−.162	−.213 [−.474]	no
Entropy	[9]	.117	11/25	+.102	—	no
Gradient magnitude	[9]	.104	11/25	+.129	—	no
Saliency area	[9]	.075	3/25	+.031	—	no
Averaged aesthetics (3)	[2]	.032	8/25	+.027	—	no
Composite of five	[10]	.029	11/25	−.020	+.416 [+.160]	no
GLCM contrast	[12]	.019	9/25	+.136	—	no
Laplacian variance	[1]	−.047	9/25	+.095	+.551 [+.328]	no
Saturation	[4]	−.092	7/25	−.254	−.034 [−.325]	no
Colourfulness (Hasler)	[3]	−.182	4/25	−.062	+.281 [+.011]	no
Colourfulness (Ke)	[7]	−.182	4/25	−.062	+.281 [+.012]	no

Table S18 Which operation entered which test. A filled marker means the operation has a result on that axis. **Coverage is not uniform, and the two reasons differ.** Spatial information carries no paired verdict because its source declares no direction, which is a property of the source, and for the same reason it has no downstream result either. The other six operations without a downstream result were added to the implementation after the downstream pass was run, which is a property of our sequence of work and not of the method; extending that pass to them is a stated next step. Sensitivity and generalization cover all 16. Counts per axis are in the last row.

Operation	Paired	Sens.	Gen.	Floor	Down.	Absent because
Averaged aesthetics (3)	●	●	●	○	○	after the downstream pass
Colourfulness (Hasler)	●	●	●	●	●	
Colourfulness (Ke)	●	●	●	●	●	
Composite of five	●	●	●	●	●	
Contrast	●	●	●	●	●	
Edge distribution	●	●	●	●	●	
Entropy	●	●	●	○	○	after the downstream pass
GLCM contrast	●	●	●	○	○	after the downstream pass
Gradient magnitude	●	●	●	○	○	after the downstream pass
Gradient orientation	●	●	●	○	○	after the downstream pass
Grey-level dispersion	●	●	●	●	●	
Hue count	●	●	●	●	●	
Laplacian variance	●	●	●	●	●	
Saliency area	●	●	●	○	○	after the downstream pass
Saturation	●	●	●	●	●	
Spatial information	○	●	●	●	○	source declares no direction
of 16	**15**	**16**	**16**	**10**	**9**	

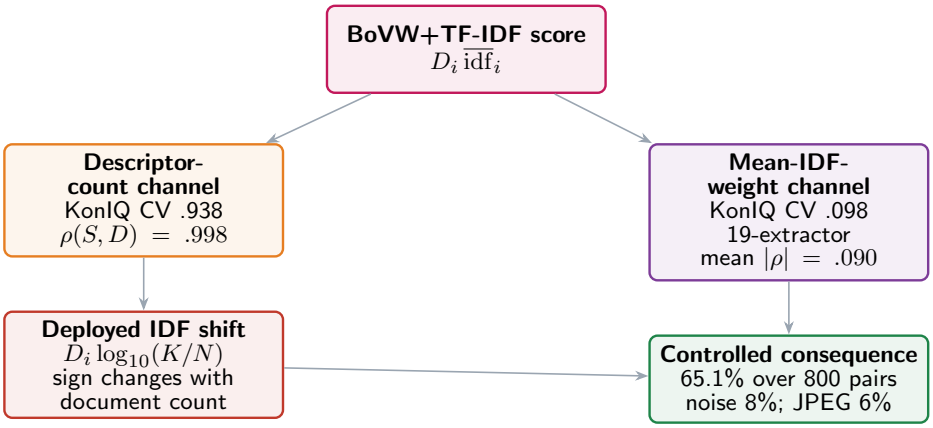


Fig. S1 Measured mechanism. The factorization identifies two channels; the data determine their relative variation and connect count dominance to the controlled-degradation failures.

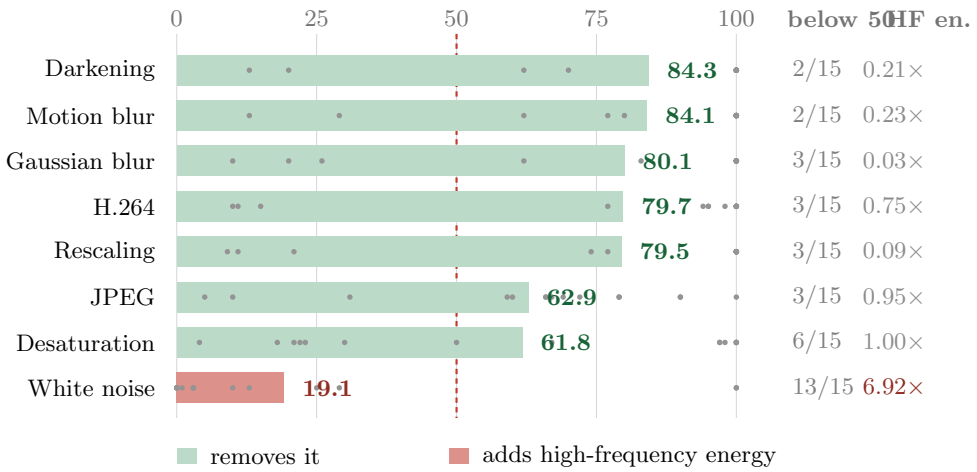


Fig. S2 Mean paired accuracy of the fifteen summarization operations, by degradation type; grey dots are the individual operations and the dashed line is chance. The ordering follows what each degradation does to local image high-frequency energy, measured as the median ratio to the original and shown in the last column, rather than how severe the degradation is. Seven of the eight reduce that energy and are answered correctly. **White noise is the only one that raises it**, by a factor of 6.9, and it is answered at 19.1 per cent with a median of 3.0. The middle column counts operations below chance.

Table S19 Model-reference output over the 42 operational redundancy clusters. Changed counts representatives differing from the deployed operation. DOVER-T is the raw DOVER technical branch and is not used in screening; Q-Align defines the in-domain screening floor. Both outputs use the global-achievable-range estimand of Eq. 3 of the main article; Max is the percentage of clusters in which the selected representative has the highest DOVER-T score.

Quality operation	Changed	DOVER-T	Max	Q-Align
SIFT+BoVW	—	50.5%	54.8%	48.6%
LIQE	**22/42**	**76.0%**	59.5%	90.2%
QualiCLIP	19/42	71.9%	59.5%	80.6%
CLIP-IQA+	22/42	70.7%	50.0%	74.3%

Table S20 Complete-case method screening. Sensitivity reports the within-reference mean and number of KADID types above .50. Generalization is the lowest signed ρ^* over all unseen databases. Floor is normalized agreement with Q-Align on the common 42 clusters and its topic-block lower bound (L). Sup.: K = KonIQ-10k opinion training; L = LIVE; OU = opinion-unaware; – = no opinion training; M = multi-database mixture. A dagger marks a value measured on a database the method trained on: Q-Align's mixture includes KADID-10k. Q-Align is a reference, not a candidate.

Method	Sup.	Pol.	Sensitivity	Lowest unseen ρ^*	Floor; L	Eligible
LIQE	K	↑	.807; 22	.723	.551; .284	**yes**
QualiCLIP	OU	↑	.849; 22	.665	.371; .143	**yes**
CLIP-IQA+	K	↑	.808; 22	.654	.236; .034	**yes**
TOPIQ	K	↑	.721; 20	.511	.506; .289	no
MUSIQ	K	↑	.733; 20	.556	.551; .297	no
DBCNN	K	↑	.668; 19	.484	.461; .274	no
HyperIQA	K	↑	.673; 19	.462	.371; .122	no
NIQE	OU	↓	.607; 19	.368	.281; .012	no
BRISQUE	L	↓	.657; 19	.236	.371; .136	no
CLIP-IQA	OU	↑	.645; 20	.501	.371; .045	no
SIFT+BoVW (deployed)	–	↑	−.077; 5	−.445	.056; −.308	no
Q-Align	M	↑	.880†; 23†	–	–	reference

Table S21 Two published expressions against the property the same paragraph claims for them. **Above**: the contrast term of the Qaroush visual-attention objective, over a sweep in which contrast rises by construction (25 levels, seed 42). Histogram equalisation drives every frame towards the same output standard deviation, so the published difference measures distance from maximum contrast: it runs backwards, and so does the signed form. Only the ratio, which the equation does not write, agrees. **Below**: the brightness term of the Korolkov module, over a sweep in which exposure rises by construction (21 levels). The source states that it "penalizes under- and over-exposed frames", which requires a non-monotone function; the mean luminance it specifies is monotone and enters the module with weight +0.3, so a brighter frame always scores higher. **No corpus is involved: both inversions are properties of the expressions.**

Reading	Expression	ρ with the property
Qaroush contrast, against contrast (Eq. 8)		
Eq. 8 as published	$\frac{1}{K}\sum_k \lvert\sigma_p(\mathrm{HE}(f_k)) - \sigma_p(f_k)\rvert$	−1.0000
without the absolute value	$\frac{1}{K}\sum_k \big(\sigma_p(\mathrm{HE}(f_k)) - \sigma_p(f_k)\big)$	−1.0000
ratio, unwritten in the source	$\frac{1}{K}\sum_k \sigma_p(f_k)/\sigma_p(\mathrm{HE}(f_k))$	+1.0000
Korolkov brightness, against exposure		
as published	$\frac{1}{\lvert I\rvert}\sum_{x,y} L(I)_{x,y}$ (LAB)	+1.0000
what "penalizes" would need	$-\lvert L - 127.5\rvert$	−.4883

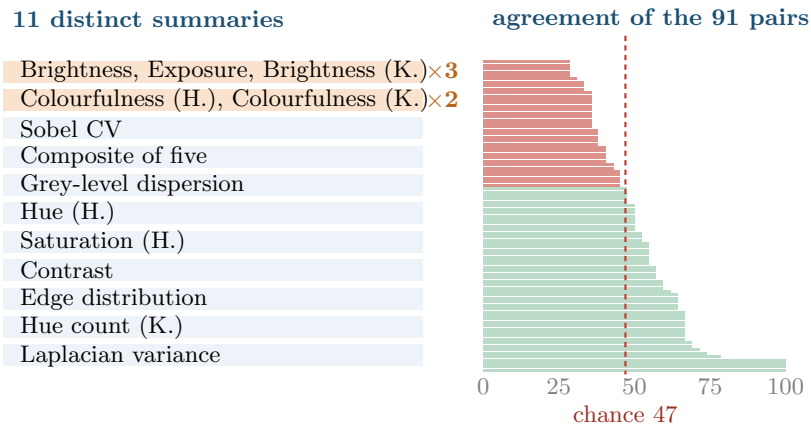


Fig. S3 The same 42 redundancy clusters, the same pipeline, only the visual-selection operation swapped; no reference model is involved. **Left**: fourteen operations produce eleven distinct summaries, and only two groups agree completely: the three brightness variants and the two colourfulness implementations. **Right**: agreement for each of the 91 operation pairs, sorted, against the chance floor of 47 per cent. Every cluster has at least two operations choosing differently, the average cluster draws 2.21 distinct choices, and **37 of the 91 pairs agree below chance**. **Pairs drawn from the same source work agree no more than pairs from different works**: median κ .034 over 30 within-work pairs against .056 over 60 across-work pairs, with the two exact duplicates excluded. The choice of operation is not a detail of the stage; it is most of what the stage decides.

Table S22 Traceability of every audited operation: the source work, the formula as the source states it, the function that computes it, and the reference model used on the in-domain column. Every operation name and count printed elsewhere in this article is derived from this table. **Dir.** is the direction the source declares; spatial information declares none, which is why it is measured but carries no verdict.

Operation	Source	Formula, as the source states it	Function	Dir.	Ref.
Averaged aesthetics (3)	[2]	three aesthetic attributes; unweighted mean‡	`gygli_media_crua`	↑	LIQE
Colourfulness (Hasler)	[3]	$M^{(3)} = \sigma_{rgyb} + .3\mu_{rgyb}$	`hasler_colorfulness`	↑	LIQE
Colourfulness (Ke)	[7]	Hasler $M^{(3)}$; Ke gives no formula	`hasler_colorfulness`	↑	LIQE
Composite of five	[10], [7]	unweighted mean of the five Ke terms	`srinivas_quality`	↑	LIQE
Contrast	[7]	width of central 98% of the grey histogram (Ke §4.5)	`ke_contraste`	↑	LIQE
Edge distribution	[7]	$q_a = 1 - w_x w_y$ (Ke §4.1)¶	`ke_edge_distribution`	↑	LIQE
Entropy	[9]	Shannon entropy over HSV histogram bins (Eq. 5)	`qaroush_entropia`	↑	LIQE
GLCM contrast	[12]	GLCM contrast descriptor	`glcm_contraste`	↑	LIQE
Gradient magnitude	[9]	mean Sobel gradient magnitude (Eq. 10)	`qaroush_gm`	↑	LIQE
Gradient orientation	[9]	SD of Sobel gradient orientations (Eq. 15)	`qaroush_go`	↑	LIQE
Grey-level dispersion	[6], [11]	grey-level standard deviation§	`contraste_variancia`	↑	LIQE
Hue count	[7]	$q_h = 20 - \lvert N \rvert$, $\alpha = .05$ (Ke §4.3)	`ke_hue_count`	↑	LIQE
Laplacian variance	[1], [8]	variance of the grey-scale Laplacian	`laplacian_var`	↑	LIQE
Saliency area	[9]	area fraction above a saliency threshold (Eq. 6)	`qaroush_saliencia`	↑	LIQE
Saturation	[4]	mean HSV saturation (§III-A-1)	`huang_saturacao`	↑	LIQE
Spatial information	[5]	Sobel energy, ITU-T P.910	`si`	none	LIQE

All sixteen are computed by `ops_da_literatura.py` on the same frames and pooled identically. **Ref.** is the model each operation is compared against on the in-domain column; the ten no-reference models in the main screening table use Q-Align there instead, so that column compares each family with its own reference. Colourfulness has no formula in Ke, who cites Datta; the Hasler expression is used and the substitution is declared here rather than silently. Scores for fifteen operations are in `corpus_inteiro.json`; spatial information is scored on the extended 83-segment design by `si_no_corpus_de_video`, in `designA_estendido_83.json`.

‡ Gygli et al. [2] lists the attributes but not their combination; **the unweighted mean is a choice made here.** ¶ Ke builds the edge map with a Laplacian per RGB channel at 100×100; here it is Sobel magnitude on the grey frame, so **the formula is reproduced and the map adapted**. With the source map sensitivity goes 0.298→0.272 and types 9→7 (versions correlate 0.876); no verdict changes. § Jeong et al. [6] states *variance of pixels* and Xu et al. [11] *standard deviation of the grey values*; the the standard deviation is implemented. **Within a frame the two order alike; a mean over frames does not**: under the mean of variances 7 of the 800 pairs move and no per-type verdict does.

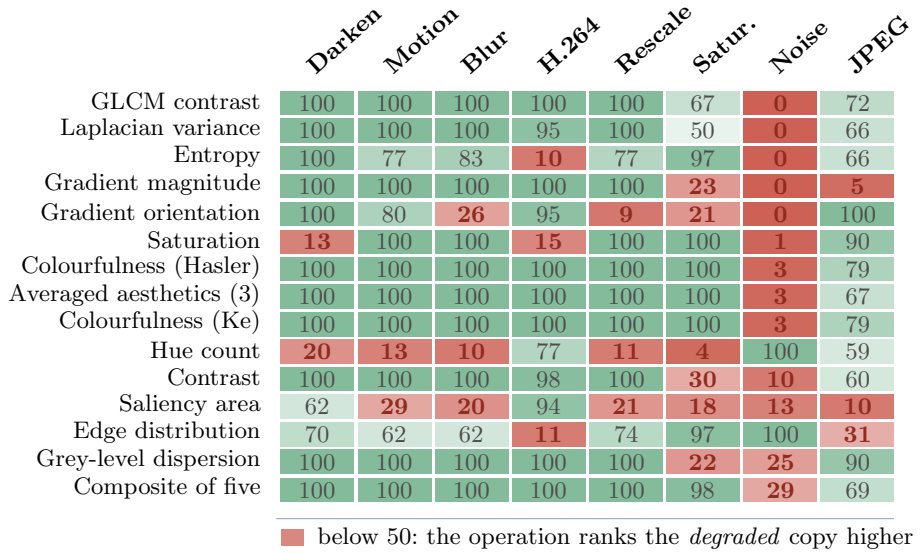


Fig. S4 Paired accuracy of each summarization heuristic on 800 unmodified–degraded broadcast pairs. Each cell is 100 pairs, 20 segments at five severity levels, and the better member is fixed by construction. **Every operation falls below chance on at least one type, and thirteen of the fifteen do so under white noise**, five of them on every pair. Rows are ordered by worst cell.

Table S23 The three families under the screening rule of Section 3.5 of the main text, whose three conditions are stated there. **Each cell is the maximum over its family on that axis alone**, and the two maxima need not belong to the same member: among the no-reference models the best sensitivity is QualiCLIP and the best generalization LIQE. **The composite row is screened under a different orientation**: a combination has no direction declared by any source, so its sensitivity is oriented by the sign of the data rather than by a stated direction (Table S30). That gate is the more permissive of the two, which makes the zero in its eligibility column conservative. **Among the composites the same split is decisive**: the best generalization belongs to a combination that fails the sensitivity condition, while the single composite that clears sensitivity generalizes at only .061. Spatial information declares no direction and the reproduced modules were measured on the constructed pairs rather than on these axes, so neither carries a verdict here; the heuristics row counts 16 computed and 15 evaluated. Per-operation values are in Table S18 and per-model values in Table S21.

Family	n	Best sens.	Best gen.	Clears both
Summarization heuristics	16	+.298	+.139	0 of 15 evaluated
Published composites	19	+.524	+.601	0 of 19 (1 clears sens.)
Reproduced modules	2	—	—	not screened
No-reference models	10	+.849	+.723	**3 of 10**
Deployed SIFT+BoVW	1	−.077	−.445	0 of 1

Table S24 Origin, documented use and evaluated object for the cases that carry the claims here. **A citation does not say which work used the operation in selection, nor what this bench measured**, so the three are separated; where origin and use fall in the same work the middle column says so, and the last column marks our own adaptations as such. The full inventory, with formulas and results files, is in Tables S17 and S23.

Operation	Origin and role there	Documented use in selection	Evaluated here
Edge distribution, hue count, contrast	Ke et al. [7]: features of a naive Bayes classifier for photo quality	Srinivas et al. [10] cites Ke for all three (§3.1.3–3.1.5) and averages them into a per-frame quality score	each term alone, plus the mean and an adapted classifier
Colourfulness in that same score	Swain and Ballard, by histogram intersection, as Srinivas et al. [10] states	Srinivas et al. [10], §3.1.1	the Hasler expression instead, a declared substitution
Sharpness and brightness	Korolkov [8]: two terms of a keyframe score	the same work, $0.7\,z(\text{sharp}) + 0.3\,z(\text{bright})$	the published rule, with our normalization population stated
Grey-level dispersion	Jeong et al. [6]: a badness term inside a cost multiplicative in cluster distance	the same work, within that objective	the isolated statistic, read in the increasing direction
Gradient magnitude, saliency, entropy, gradient orientation	Qaroush et al. [9]: terms of four numbered objectives	the same work; two of the four objectives are not about technical quality	each term alone, and the visual-attention module of Eq. 20

Table S25 What each source work reports as evidence for its own visual criterion. Read from the full text of each paper; the excerpt is verbatim. *Tests* names what the source puts to a test, which is not always the operation the later literature borrows. One of the eleven source works is not in our archive and carries no verdict here: **absence of a reading is not absence of validation**. It documents the clause carried on each band of the audit table in the main text.

Source	Tests	What it reports	Verbatim excerpt
Ke et al. [7]	**human judgment of the property**	community ratings on 60,000 photos as ground truth; per-feature precision–recall and a combined classifier at 27.8 per cent equal error, on the extremes only	*"the photos have been rated by its community of users, giving us ground truth on the quality of each photo . . . The top and bottom 10% of the photos were extracted . . . we chose to ignore the middle 80%"*
Hasler and Süsstrunk [3]	**human judgment of the property**	a psychophysical scaling experiment, seven categories, the expression fitted to the ratings at 94.0 per cent; the property is colourfulness	*"We set up a psychophysical category scaling experiment, and ask people to rate images using 7 categories of colourfulness. We then fit a metric to the results"*
Srinivas et al. [10]	the summary only	precision, recall and F of the selected keyframes against the ground-truth keyframes of seven open-video.org videos	*"results are compared with the ground truth keyframes provided by open-video.org"*
Qaroush et al. [9]	the summary only	precision, recall and F1 of the extracted segments against reference summaries	*"The evaluation focuses on comparing segments extracted by our system with those identified in ground truth summaries"*
Korolkov [8]	**neither the rule nor its weights**	weights set, not fitted; ablations vary segmentation parameters, and keyframes are judged by coverage and a qualitative review	*"We set wsharp = 0.7, wbright = 0.3 to favor focused frames"*

Table S26 Evidence reported by each source work, continued. Columns as in the preceding table.

Source	Tests	What it reports	Verbatim excerpt
Cakmak et al. [1]	the summary, with an ablation	TVSum20 and SumMe against human importance scores, with an ablation removing each modality	*"Removing any single component"*
Yang et al. [12]	the summary, with an ablation	structural and feature ablations scored by F on SumMe over 15 annotations	*"we conducted experiments on the SumMe dataset for structural ablation of the model and for feature ablation"*
Gygli et al. [2]	the summary, with an ablation	f-measure against multiple human-created summaries per video, collected in a study with perception psychologists, plus an ablation measuring the gain of adding each feature to the set	*"We evaluate our method using multiple 'ground truth' summaries per video, which we acquired in a study in collaboration with perception psychologists"*
Jeong et al. [6]	the summary only	mean opinion scores from human subjects and precision/recall against labeled events; no ablation or isolated test of the contrast term	*"the visual quality is the combination of contrast, blur, and image skew measures"*
Huang et al. [4]	the summary only	Spearman and Kendall rank correlation against human annotations on TVSum and SumMe	*"rank correlation coefficients measure how well the ordering"*

Table S27 Resampling unit per analysis, and the rival unit tested against it. Observations are nested, so the unit decides the width of every interval: the italic line under each row states the nesting, and **the declared unit follows from that nesting**, observations sharing content or a source entering the same block. Each interval was then recomputed under the rival unit; **that check identifies which implementation produced a published bound**, and is reported as a verification rather than as the reason for the choice. One of the five did not pass and was corrected; the last row is the contrast that previously declared no unit at all.

Block	Analysis	Declared unit	Rival tested	Outcome
Block 01	Paired difference, 800 pairs	segment block	the pair	**the published interval matched the rival scheme; corrected to** $[+29.8, +38.9]$
	20 segments, 40 pairs each; pairs within a segment share content			
Block 01	Strict accuracy, 800 pairs	topic block	the segment block	reproduces under the declared unit and not under the rival
	the 20 segments come from 13 news topics			
Block 04	In-domain floor, 42 clusters	topic block	the cluster	reproduces under the declared unit and not under the rival
	the 42 clusters come from 19 topics			
Block 04	Output contrast, 42 clusters	topic block	the cluster	reproduces under the declared unit and not under the rival
	the 42 clusters come from 19 topics			
Block 04	Regime contrast, 42 clusters	topic block	the cluster (exact McNemar)	the verdict holds under the declared unit ($p = .008$ against $p = .006$)
	the 12 discordant pairs fall in 9 topics; the null enumerates all 2^9 *block sign assignments*			

Table S28 Every operation–type cell below the 50 per cent chance floor, with the accuracy of each. Significance is an exact sign-flip randomization over the 20 broadcast segments, the unit the design implies, with Holm across the 120 operation–type tests: 22 of the 35 cells survive it and † marks the 13 that do not. Counting ties inside the denominator instead of outside leaves the same 35 cells and the same 15 operations. Removing the two types aligned with an operation's own attribute, darkening and desaturation, leaves all 15 operations still inverting; removing white noise instead leaves 9. Dropping one segment at a time, all 15 keep inverting in every one of the 20 rounds. **Ties are unevenly spread**: 4 operations tie on at least one pair and the largest count is 81. Where ties are many, accuracy below 50 per cent does not by itself mean the degraded copy is preferred; the reading that does is correct against incorrect, which is what the sign-flip test compares. Dk = darkening, Mo = motion blur, Bl = Gaussian blur, H2 = H.264, Rs = rescaling, St = desaturation, Ns = white noise, Jp = JPEG.

Operation	Types	Worst	Cells below chance (%)
Saliency area	6	10.0	Mo 29.0†, Bl 20.0†, Rs 21.0†, St 18.0, Ns 13.1, Jp 10.0
Hue count (Ke)	5	4.2	Dk 21.7†, Mo 15.7†, Bl 12.3, Rs 14.1†, St 4.2
Gradient orientation	4	0.0	Bl 26.0†, Rs 9.0, St 21.0†, Ns 0.0
Gradient magnitude	3	0.0	St 23.0†, Ns 0.0, Jp 5.0
Saturation	3	1.0	Dk 13.0, H2 15.0, Ns 1.0
Entropy	2	0.0	H2 10.0, Ns 0.0
Contrast (Ke)	2	10.3	St 31.6†, Ns 10.3
Edge distribution	2	11.0	H2 11.0, Jp 31.0†
Grey-level dispersion	2	22.0	St 22.0†, Ns 25.0†
GLCM contrast	1	0.0	Ns 0.0
Laplacian variance	1	0.0	Ns 0.0
Gygli mean of three	1	3.0	Ns 3.0
Colourfulness (Hasler)	1	3.0	Ns 3.0
Colourfulness (Ke)	1	3.0	Ns 3.0
Srinivas composite	1	29.0	Ns 29.0

Table S29 The deployed operation measured twice on the same 800 pairs: the reported run and an independent reimplementation of the same published operation under the same protocol. **The failure reproduces**: both collapse on white noise and JPEG, and both stay below the weakest published no-reference model at 80.8 per cent. They differ on desaturation, where the reported run inverts and the reimplementation does not, which is the one place where implementation detail changes a verdict. Red marks a cell below the 50 per cent chance floor.

Run	Darken	Motion	Blur	H.264	Rescale	Satur.	Noise	JPEG	Mean
Reported	100	100	90	90	88	**39**	**8**	**6**	**65.1**
Independent reimplementation	100	100	92	94	93	56	**10**	**6**	**68.9**

Table S30 The 19 attribute combinations the summarization literature publishes, with the number of archive papers that use each. These are not ensembles assembled here: each row is a set of terms some published work combines. *Sens.* is the within-reference mean against the degradation level on KADID-10k under the z-score aggregation. **Orientation here comes from the data**, the sign of the mean across types, because a combination of attributes has no direction declared by any source, whereas Table S18 orients each operation by the direction its own source states. Magnitudes match across the two tables; a single-attribute row therefore carries the opposite sign to Table S18 exactly when the declared direction contradicts the data. **This makes the gate here the more permissive of the two**, since $|\rho| \geq \rho^*$, so the verdict below is conservative: no composite clears it even under the looser orientation. *Types* is the count above .50 of 25; *Gen.* is the direction-aligned correlation on KonIQ-10k as a magnitude, under the same orientation rule as the sensitivity column. **Sensitivity and Gen. (z) are the same configuration**; the ridge column is the generous variant, fitted outside the reference and reported separately rather than mixed into the verdict. **Eligibility below uses the z-score configuration throughout**, and the count is the same under either consistent choice: zero of the 19 are eligible whether both gates use the z form or both use the ridge form. **1 of the 19 clears the sensitivity gate** (green) and none is eligible, because that one does not transfer.

Combination	Papers	Sens.	Types	Gen. (z)	Gen. (ridge)	Eligible
colourfulness + Laplacian variance + LBP uniformity	1	**.524**	17	.061	.067	no
Laplacian variance + LBP uniformity	1	.439	15	.080	.082	no
edge distribution	2	.298	9	.099	.099	no
spatial information	5	.251	13	.513	.513	no
LBP uniformity + gradient magnitude	1	.189	8	.177	.007	no
colourfulness	2	.182	9	.007	.007	no
LBP uniformity	4	.175	12	.205	.205	no
colourfulness + exposure	1	.173	10	.243	.003	no
edge distribution + hue count + Laplacian variance	1	.156	7	.204	.347	no
colourfulness + edge distribution + hue count + saliency	2	.148	7	.162	.117	no
exposure + Laplacian variance	1	.108	9	.500	.357	no
colourfulness + gradient magnitude	1	.108	11	.296	.066	no
exposure	1	.102	4	.343	.343	no
GLCM contrast + entropy	1	.092	11	.371	.084	no
LBP uniformity + saliency	3	.089	9	.071	.277	no
saliency	6	.075	3	.333	.333	no
Laplacian variance	3	.047	9	.601	.601	no
colourfulness + edge distribution	1	.017	4	.085	.051	no
LBP uniformity + entropy	1	.009	8	.007	.285	no

Table S31 How the audited operations were identified, and what the protocol does and does not reach. The operations were not assembled from a database query, so the population of inference is the archive named in the first row and every count in the article is a count within it; no claim of exhaustive coverage is made. **The search instrument is itself measured against a manual reading, with the direction of the residual bias stated**, because a search whose accuracy is unknown cannot bound what it missed.

Step	What was done	What it yielded
Corpus	Curated full-text archive on video and multi-document summarization and on image-quality assessment, searched on 5 September 2026. **Not a database query, so the population of inference is this archive.**	186 PDFs
Patterns	One specific expression per operation family, never a bare word, so that *contrast* alone does not match. Reference lists are cut before matching; a cut leaving under 40 % of the PDF is refused and the work enters whole. Spatial information counts only when its context window carries ITU, the SI/TI pair, *index* or Sobel.	13 patterns
Instrument accuracy	Measured against a manual reading of the sources. **The bias runs in favour of the instrument**: the same author did the reading and the inventory, so recall is an upper estimate. Cutting reference lists raised precision from .47 without moving recall.	recall .81, precision .63, on 21 articles
Screening criterion	An operation enters only if its source declares a direction for its own statistic and scores a single frame. **No operation without a declared direction was screened in.** The 3 that declare a direction and stay out are sequence-level: `qaroush_movimento`, `qaroush_ses`, `sobel_cv`.	15 of 24 mapped operations
Coverage achieved	Share of the operation mentions in the archive that the screened set covers. A mention is not an article: one PDF may name several.	41 of 71 mentions (57.7 %)
What is left out, and why	**Of the 30 mentions outside, 25 are not screenable by any rule**, because the source declares no direction (14) or gives no combination rule (11). That residue is under-specification in the sources, and it is the same property that would stop an independent reader from reimplementing them.	25 of 30 not screenable
What the archive does instead	Works citing no dedicated no-reference quality model, and how many of those compute one of the audited operations in its place. **A floor**: a work that computes an operation without naming it is not counted.	173 of 186 cite none; 33 of those use an operation

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